Transport of compatible panel sets

Human Compression project

Let $B \leq G$, let $\mathcal{P}_B = (P_1, \ldots, P_n)$ be the acquired complete ordered panel inventory, and let A be the image of $N_G(B)$ on its indices. The acquired rank-two test defines a symmetric relation $\sim$ on distinct indices. Conjugation preserves the subgroup and coset conditions in that test, so $\sim$ is A-invariant. Put $R(i) = \{j : i \sim j\}$.

Lemma 1. *Choose one representative r from each A-orbit. Testing one representative of each A_r-orbit determines $R(r)$ exactly. If $r^a = i$, then $R(i) = R(r)^a$. This formula is independent of the chosen transporter a.*

Proof. For $s \in A_r$, invariance gives $r \sim j$ if and only if $r \sim j^s$. Thus every point stabilizer orbit is either wholly accepted or wholly rejected. The singleton orbit $\{r\}$ is rejected by the distinctness convention. This proves the first assertion. Applying the same invariance with an element a sending r to i proves the second. Any two such transporters differ by an element of A_r, which preserves $R(r)$. □

For a selected family of indices T, the complete set used in the previous generation bound is consequently

$$C(T) = T \cup \bigcap_{i \in T} R(i).$$

The intersection over an empty family means all panel indices. A panel outside T belongs to this set precisely when it is compatible with every selected panel. As before, every admissible extension of T generates a subgroup of $J(T) = \langle B, P_j : j \in C(T) \rangle$. Thus $J(T) < G$ rules out every full-group extension. This is exactly the previous bound, with the same acceptance and equivariance proof.

The implementation stores the stabilizer orbits and the test result on each representative for each chosen r. It constructs other rows only when requested, using an actual permutation $a \in A$ with $r^a = i$. It records the transporter and the complete transported index set. No quadratic compatibility matrix is required. Membership in the actual action image, the transporter equation, and the equality of the transported set are explicit certificate checks. A rejected prefix still records its exact $C(T)$ and generators of its proper join.

The continuation differs from the previously reviewed compatible-panel continuation only in these row lookups and their serialization. The original rank-two, maximal augmentation, family acceptance and equivalence tests remain unchanged. A saved complete old level is still an acquired input. Subsequent level sizes count the filtered search, while all full-group families remain covered by the previous preservation argument. Joining the panels may stop once the join equals G, since every remaining panel is already a subgroup of G.

These are elementary orbit and transport arguments. Their use here does not assert a new classification. Complete panel inventories, the exact normalizer action, finite group operations and the acquired augmentation theorem remain explicit premises. The existing full-matrix computation provides a small independent comparison for the new representation.